

	A	B	C	D	E	F	G	H	I
1	A dice is rolled 1024 times and face six is observed 160 times, test at 7% level of								
2	significance, whether the dice is unbiased or not.								
3									
4	Solution:- Here we have								
5		Sample size(n) =	1024						
6		No. of six face (x) =	160						
7		Sample prop. (p) =	0.15625	= D6/D5					
8		Population prop. (P) =	0.16667						
9		Q =	0.83333	= 1 - D8					
10		Here we set up Hypothesis as							
11		$H_0 : P = 1/6$ i.e Dice is unbiased.							
12		$H_1 : P \neq 1/6$ i.e Dice is biased.							
13									
14		S.E.(p) =	0.01165	= SQRT(D8*D9/D5)					
15		$Z_{cal.}$ =	-0.89443	= (D7-D8)/D14					
16		$ Z_{cal.} $ =	0.89443	= ABS(D15)					
17		α =	0.07	let					
18		$Z_{tab.}$ =	1.81191	=NORMSINV(1 - D17/2)					
19									
20		Decision:- Since, $Z_{cal.} < Z_{tab.}$, we accept H_0 and reject H_1 , with the							
21		conclusion that Dice is Unbiased.							
22									
23					Name: Hari Bashyal				
24					Roll no: 80010538				

	A	B	C	D	E	F	G	H	I
1	From the following information , state whether City A is more literate than								
2	City B or not.Test at 92% Confidence Level.								
3				For City A		For City B			
4		No. Of person		$n_1 =$	1000	$n_2 =$	800		
5		No. Of Literate		$x_1 =$	600	$x_2 =$	500		
6									
7									
8	Solution: - Here we have								
9				For City A		For City B			
10		No. Of person		$n_1 =$	1000	$n_2 =$	800		
11		No. Of Literate		$x_1 =$	600	$x_2 =$	500		
12				$p_1 =$	0.6	$p_2 =$	0.625		
13		Combined prop. (P) =			0.611111				
14				Q =	0.388889				
15		Here , we set up Hypothesis as							
16		$H_0 : P_1 = P_2$ i.e There is no significant difference in literacy rate of two cities.							
17		$H_0 : P_1 > P_2$ i.e City A has more literacy than city B.							
18									
19		Under H_0 ,test statistics							
20		S.E($p_1 - p_2$) =			0.023124	=SQRT(E13*E14*(1/E10+1/G10))			
21				$Z_{cal} =$	-1.081125	= (E12-G12)/E20			
22		C.I. (1 - α) =			0.92				
23		Level of significance (α) =			0.08				
24			For OTT,	$Z_{tab} =$	1.405072	= NORMSINV(1 - E23)			
25		Decision: - Since $Z_{cal} < Z_{tab}$,we accept H_0 and reject H_1 with the							
26		conclusion that City A has more literacy rate than City B.							
27									
28									
29						Name : Hari Bashyal			
30						Roll no : 80010538			